E - Commerce Recommendation pipeline

REVIEW – 3

Batch 19 - Section 4

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Abstract

E-commerce platforms provide a very large number of products, which can make it difficult for customers to find products that match their needs and interests. To solve this problem, recommendation systems study different types of user activities, such as searches, clicks, product views, ratings, cart activity, and previous purchases. These activities help the system understand what a customer is interested in. The collected information is then processed through a recommendation pipeline, where the data is collected, cleaned, prepared, and analyzed. Based on this processed data, suitable products are identified, ranked according to their relevance, and recommended to the customer. The recommendations are personalized because they are based on the individual customer's interests and behavior. This project focuses on understanding how an e-commerce recommendation system works, the different stages involved in its data pipeline, and how these personalized recommendations can improve product discovery, customer experience, and make online shopping easier and more convenient.

Introduction

- E-commerce has become a major part of everyday shopping, with platforms offering thousands or even millions of products. With so many choices available, customers may find it difficult to identify products that are relevant to their needs.

- An e-commerce recommendation system helps solve this problem by understanding customer interests and suggesting products that they are likely to prefer. It uses information such as search history, product views, clicks, ratings, cart activity, and previous purchases.

- Behind these recommendations is a recommendation pipeline. It takes user and product data, processes the information, finds relevant products, ranks them, and finally displays the most suitable recommendations to the customer.

- Therefore, recommendation systems help make online shopping more personalized, convenient, and efficient, while also helping e-commerce platforms improve customer engagement and product discovery.

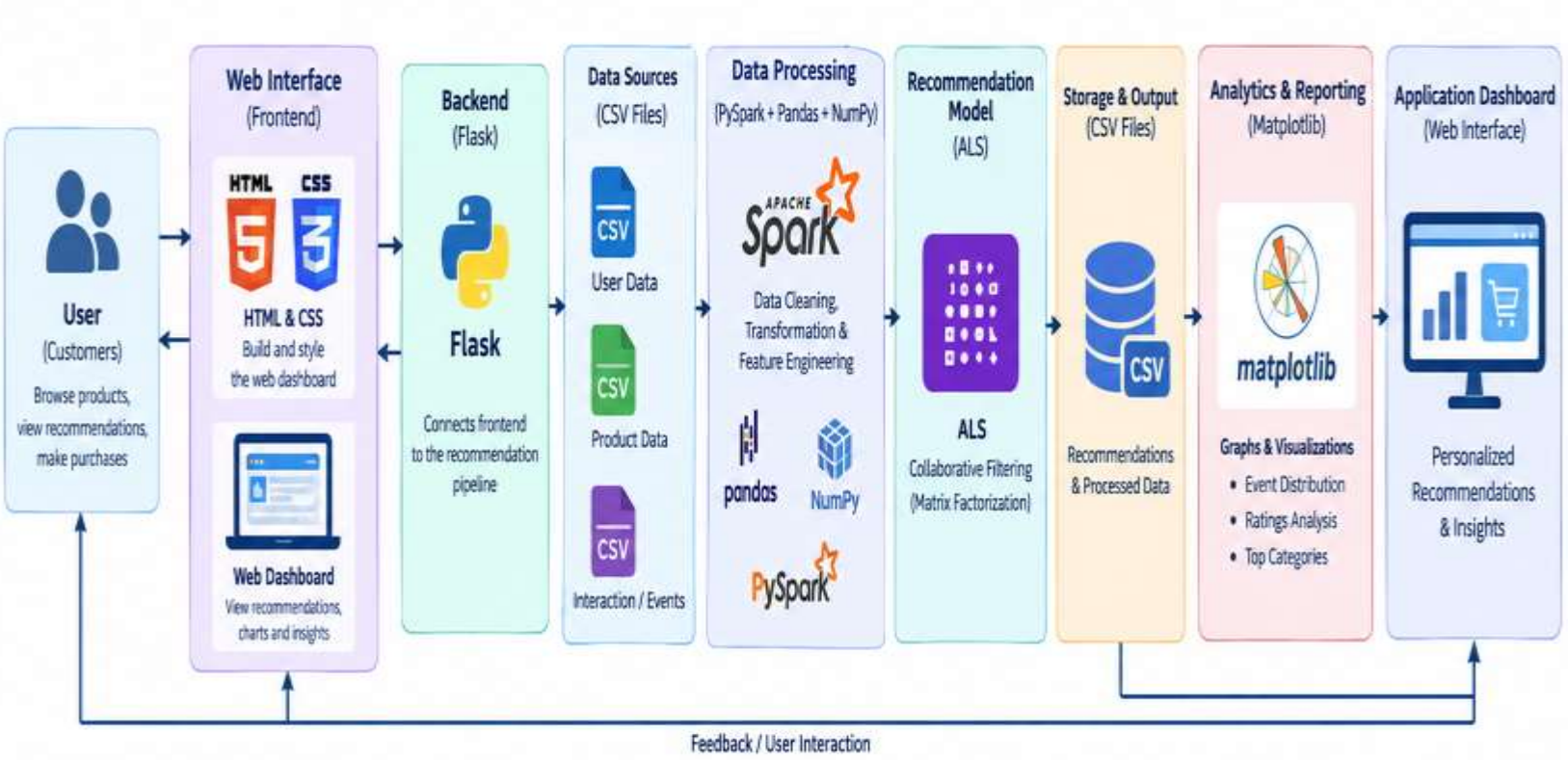
Objectives

- Generate relevant product recommendations
- Rank products based on user behavior
- Provide personalized recommendations
- Improve product discovery and customer experience

Literature Survey

S.No	Research Paper	Author(s)	**Actual Link**	Drawback
1.	Toward the Next Generation of Recommender Systems: A Survey	Gediminas Adomavicius, Alexander Tuzhilin	https://doi.org/10.1109/TKDE.2005.99	Provides a broad overview of recommendation techniques, but does not focus on one specific e-commerce application. (Wikipedia)
2.	Amazon.com Recommendations: Item-to-Item Collaborative Filtering	Greg Linden, Brent Smith, Jeremy York	https://doi.org/10.1109/MIC.2003.1167344	Works well for large-scale recommendations, but mainly focuses on item-to-item relationships and may not fully capture complex user preferences. (IEEE Xplore)
3.	Collaborative Filtering for Implicit Feedback Datasets	Yifan Hu, Yehuda Koren, Chris Volinsky	https://doi.org/10.1109/ICDM.2008.22	Uses implicit user feedback such as views and purchases, but implicit feedback does not always indicate whether a user actually liked a product. (CiNii Research)
4.	Matrix Factorization Techniques for Recommender Systems	Yehuda Koren, Robert Bell, Chris Volinsky	https://doi.org/10.1109/MC.2009.263	Provides effective recommendations, but matrix factorization can have difficulty with new users or new products due to the cold-start problem. (OA Monitor Ireland)
5.	Deep Neural Networks for YouTube Recommendations	Paul Covington, Jay Adams, Emre Sargin	https://doi.org/10.1145/2959100.2959190	Uses deep learning for candidate generation and ranking, but requires large amounts of data and computational resources. (DOI)
6.	Neural Collaborative Filtering	Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, Tat-Seng Chua	https://doi.org/10.1145/3038912.3052569	Improves recommendation using neural networks, but the model is more complex and requires more computation than traditional methods. (ACM Digital Library)

System Architecture



Methodology

Collect user and product activity data:

Gather information about user interactions, product details, views, clicks, and purchase history.

Clean and preprocess the data:

Remove missing values, duplicates, and unnecessary data. Convert the data into a suitable format for analysis.

Analyze customer interests and preferences:

Study user behavior to understand their interests and identify products they may prefer.

Generate and rank relevant product recommendations:

Use recommendation algorithms to find suitable products and rank them based on user preferences and product relevance.

Display personalized recommendations:

Present the recommended products to users through the e-commerce platform to improve product discovery and shopping experience.

Technologies and Tools

Technologies Used

1. Python

Main programming language for data processing and recommendation logic.

2. Apache Spark (PySpark)

Processes large datasets and builds the recommendation model.

3. ALS (Alternating Least Squares)

Collaborative filtering algorithm used to recommend products based on user interactions.(matrix factorization model)

4. Pandas and NumPy

Data cleaning, manipulation, and numerical computations.

5. HTML and CSS

Build and style the web dashboard.

6. Flask

Connects the backend recommendation pipeline to the web interface.

7. Matplotlib

Creates graphs for event distribution, ratings, and top product categories.

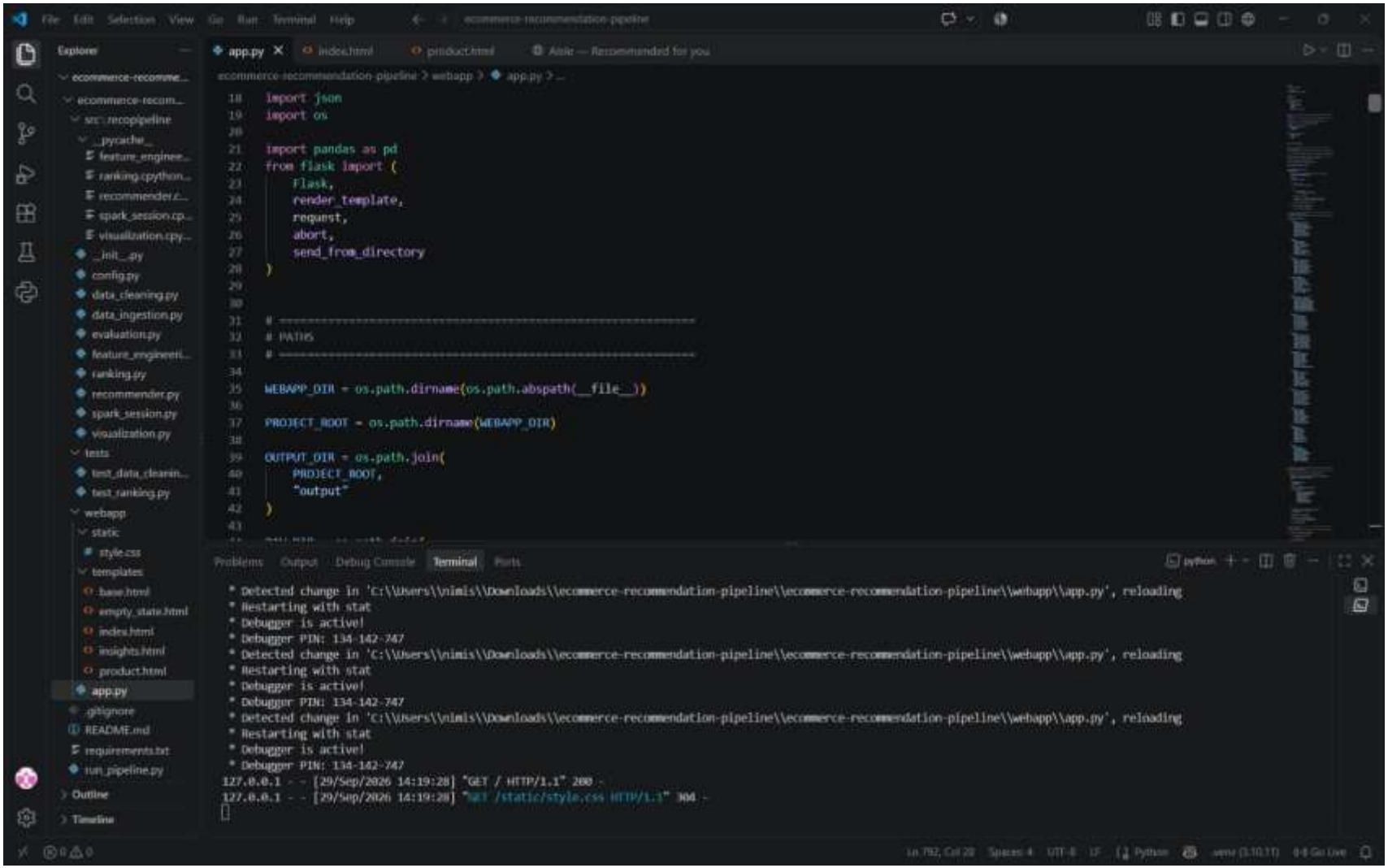
8. CSV files

Store user, product, and interaction datasets.

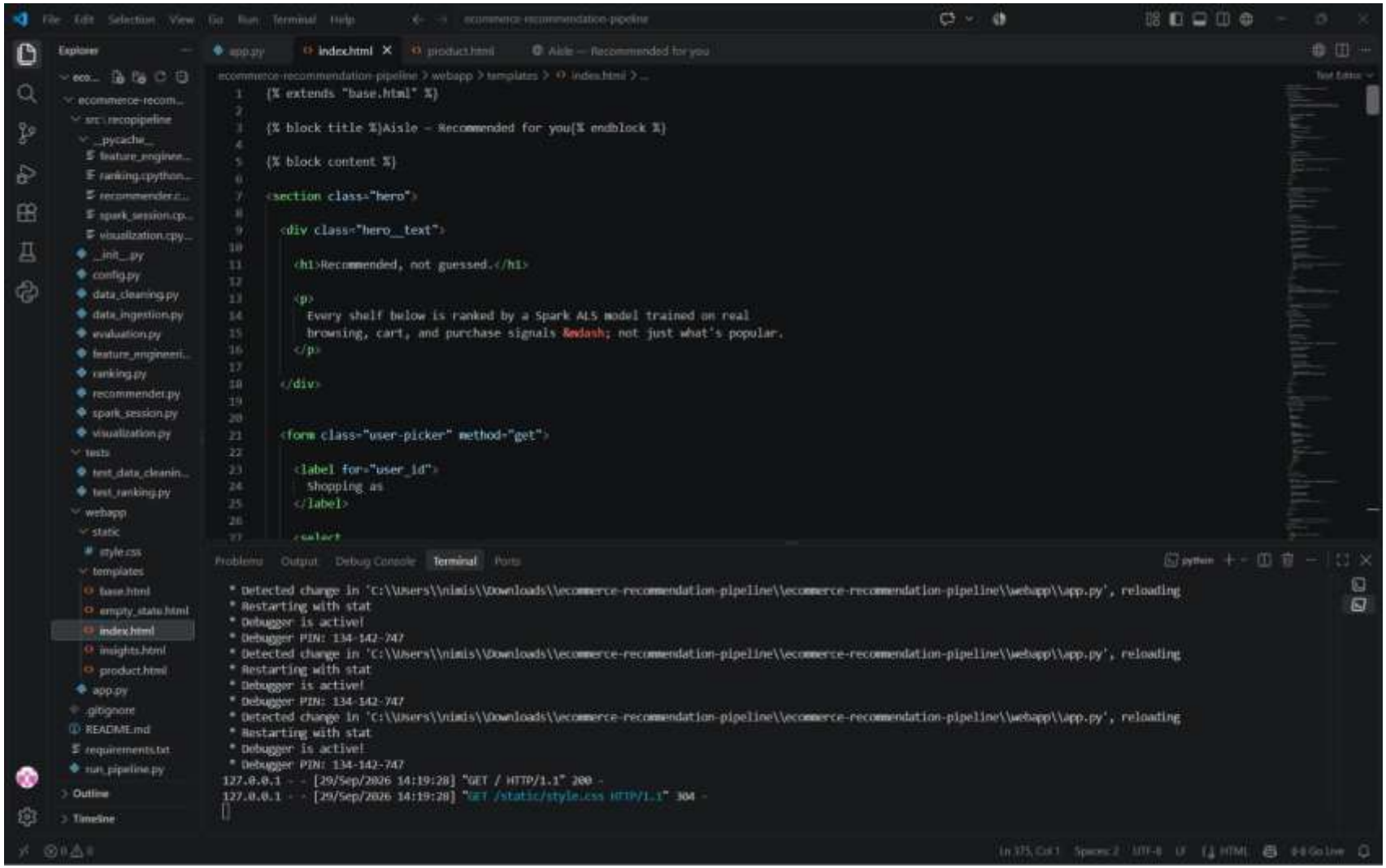
9. YAML

Stores project configuration settings.

Implementation



Implementation



Functionality

- **User Activity Collection** – The system collects user searches, clicks, product views, ratings, cart activity, and previous purchases.
- **Data Processing** – The collected user and product data is cleaned and prepared through a recommendation pipeline.
- **User Preference Analysis** – The system analyzes customer behavior to understand their interests and preferences.
- **Product Recommendation** – Based on the analyzed behavior, relevant products are identified and suggested to the user.
- **Product Ranking** – Recommended products are ranked according to their relevance to the user's behavior.
- **Personalized Recommendations** – Different users receive different product suggestions based on their individual activity.
- **Improved Product Discovery** – Users can find relevant products more easily instead of manually searching through thousands or millions of products.
- **Better Shopping Experience** – The overall goal is to make online shopping more personalized, convenient, and efficient.